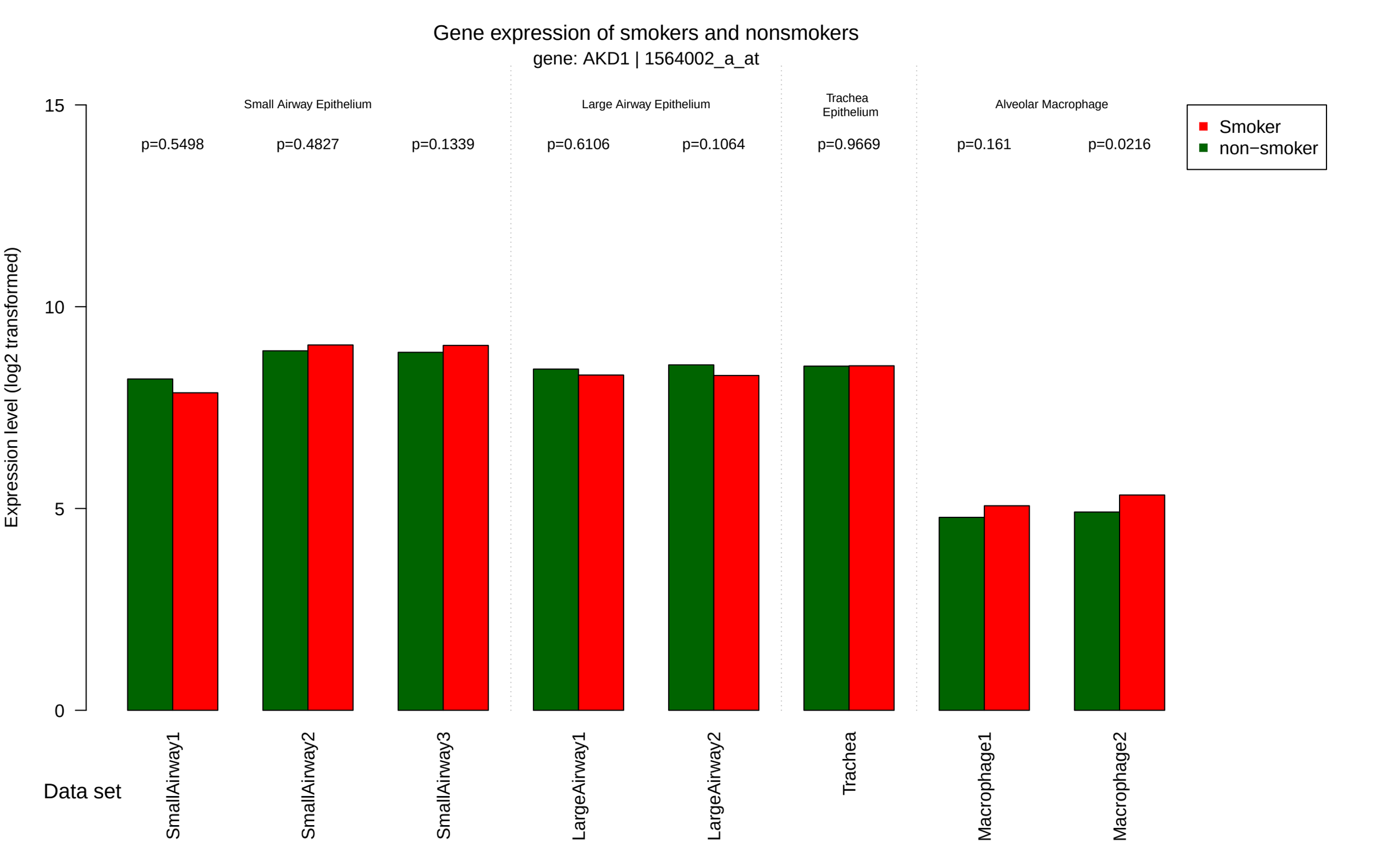
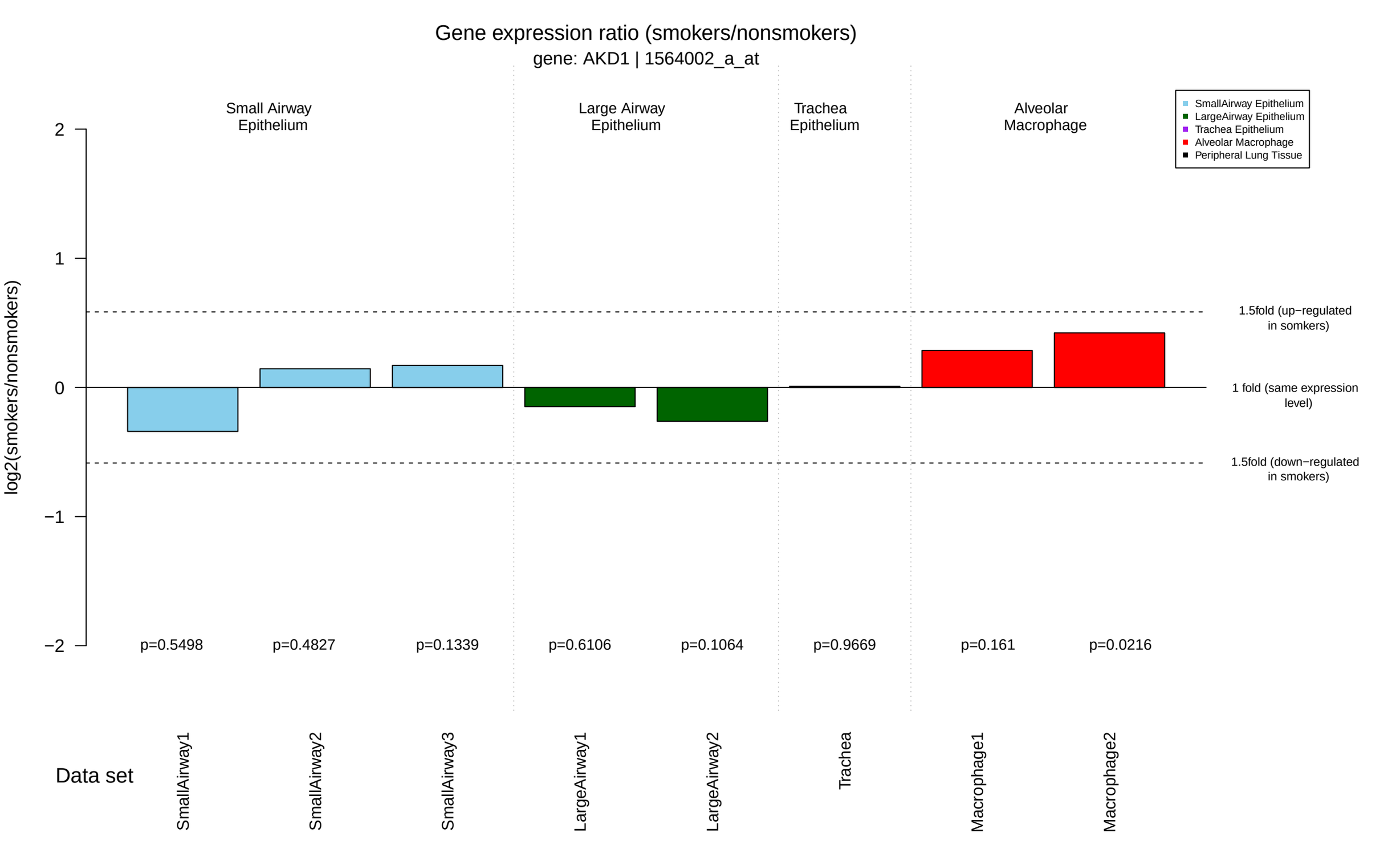
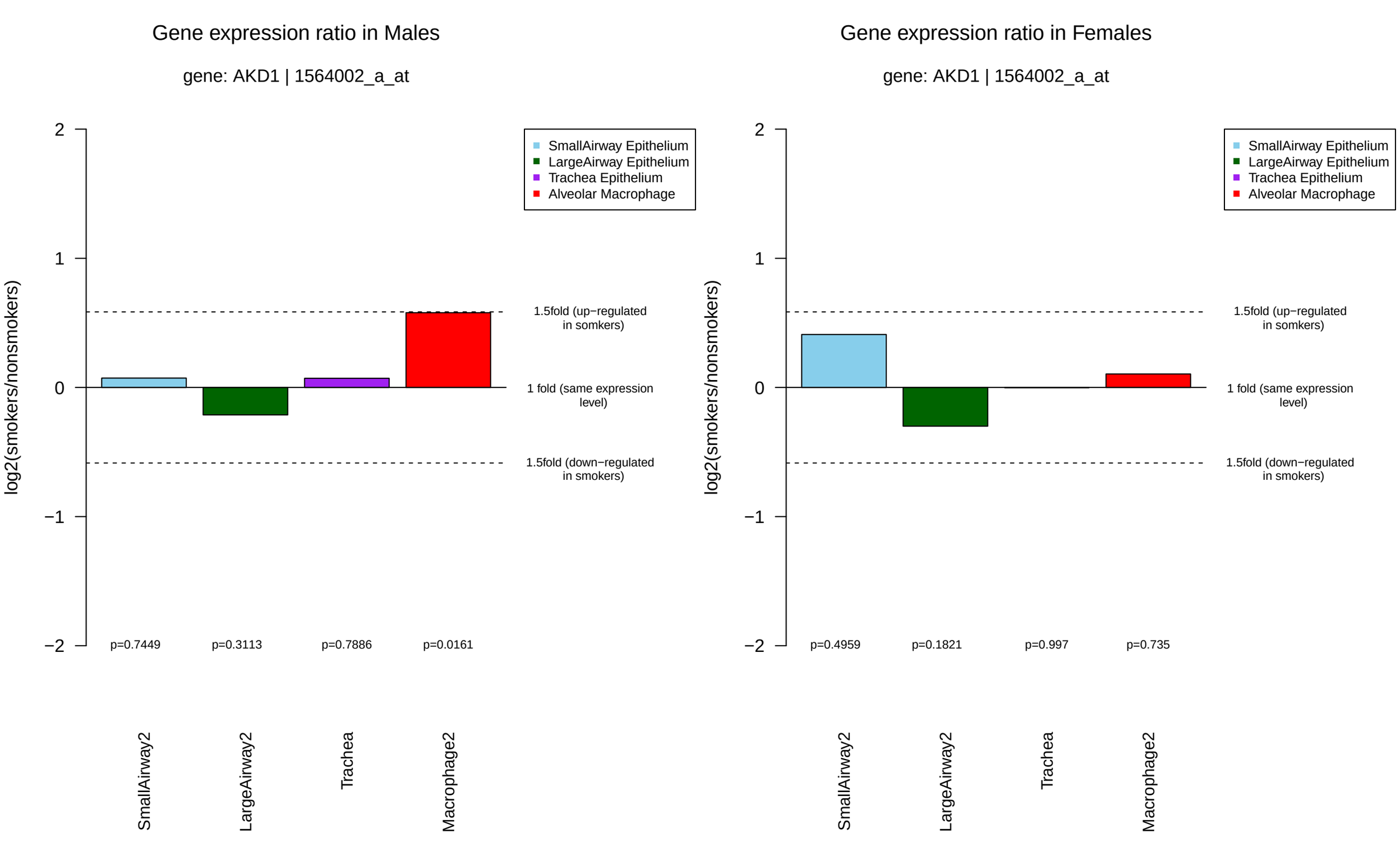
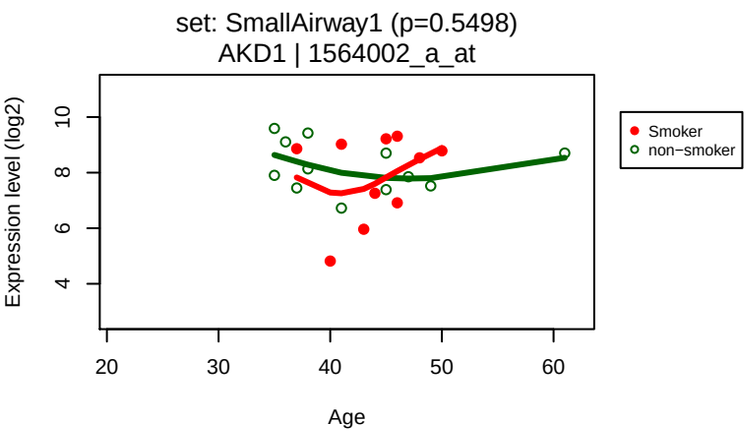
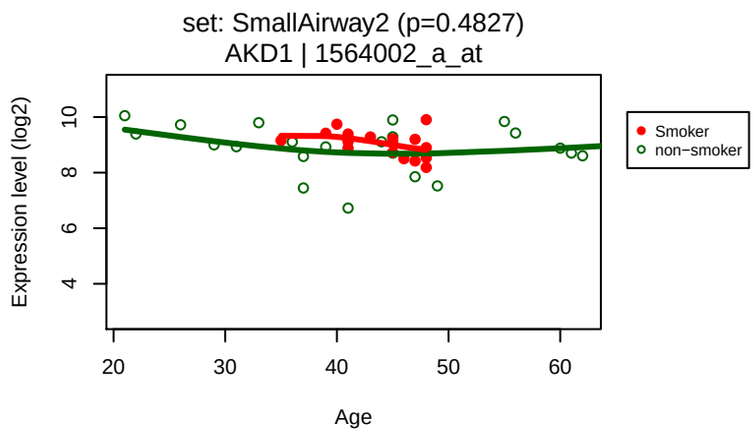
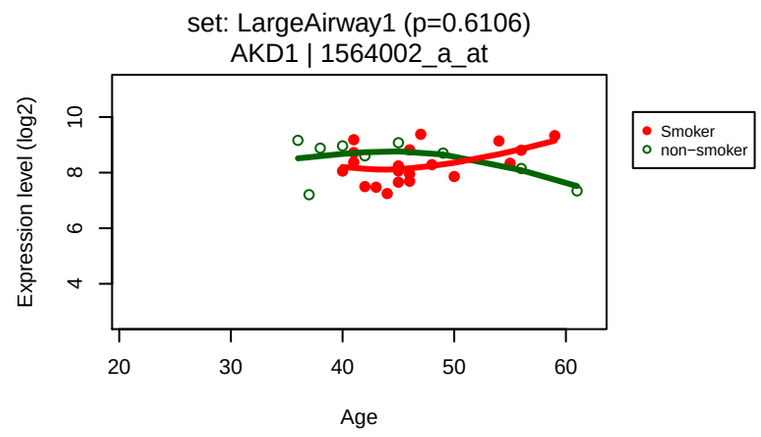

Gene expression of smokers and nonsmokers
gene: AKD1 | 1564002_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5498
p=0.4827
p=0.1339
p=0.6106
p=0.1064
p=0.9669
p=0.161
p=0.0216
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

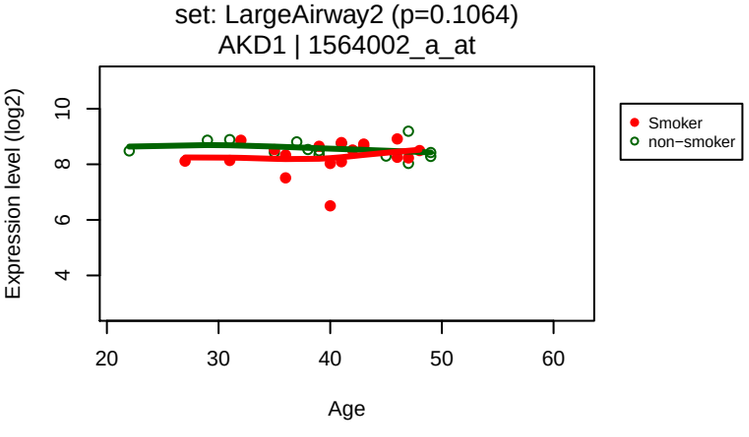

Gene expression ratio (smokers/nonsmokers)
gene: AKD1 | 1564002_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5498
p=0.4827
p=0.1339
p=0.6106
p=0.1064
p=0.9669
p=0.161
p=0.0216
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

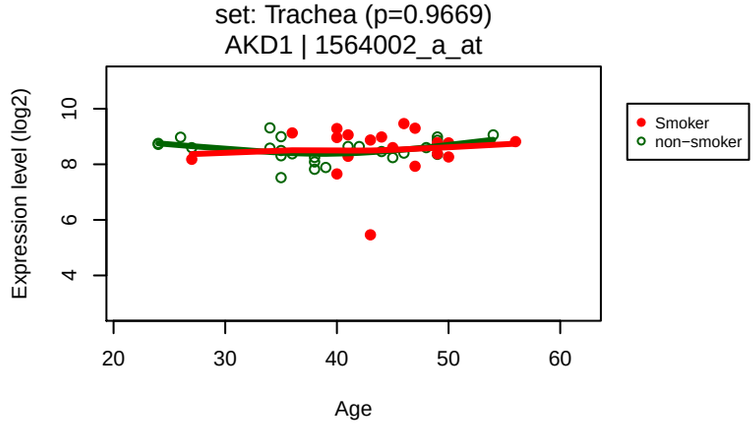

Gene expression ratio in Males
gene: AKD1 | 1564002_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7449
p=0.3113
p=0.7886
p=0.0161
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKD1 | 1564002_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4959
p=0.1821
p=0.997
p=0.735
SmallAirway2
LargeAirway2
Trachea
Macrophage2

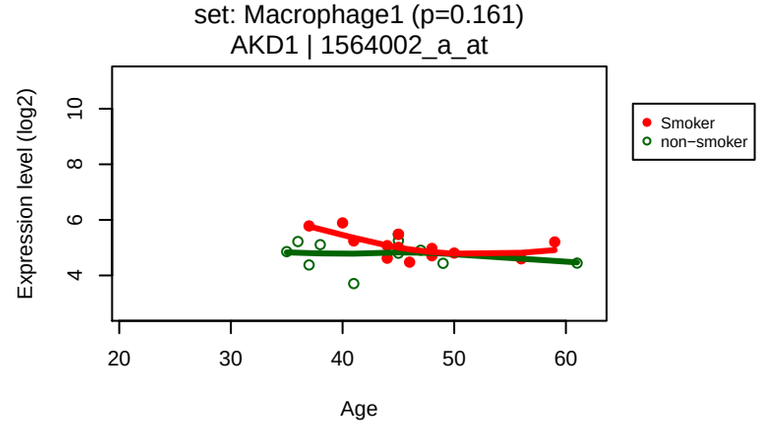

set: SmallAirway1 (p=0.5498)
AKD1 | 1564002_a_at
Expression level (log2)
Age
Smoker
non−smoker

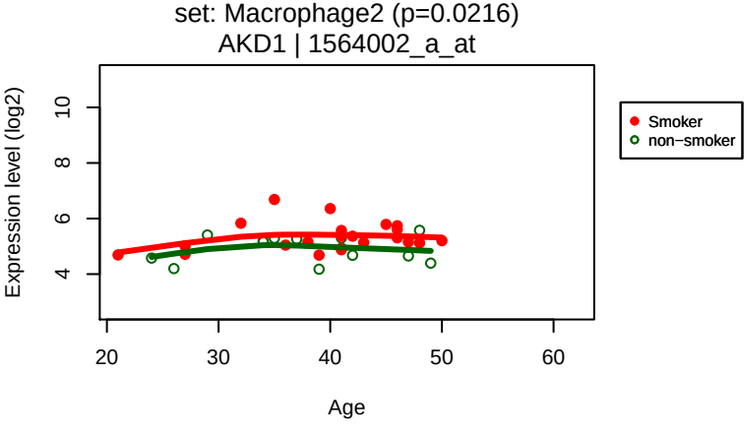

set: SmallAirway2 (p=0.4827)
AKD1 | 1564002_a_at
Expression level (log2)
Age
Smoker
non−smoker

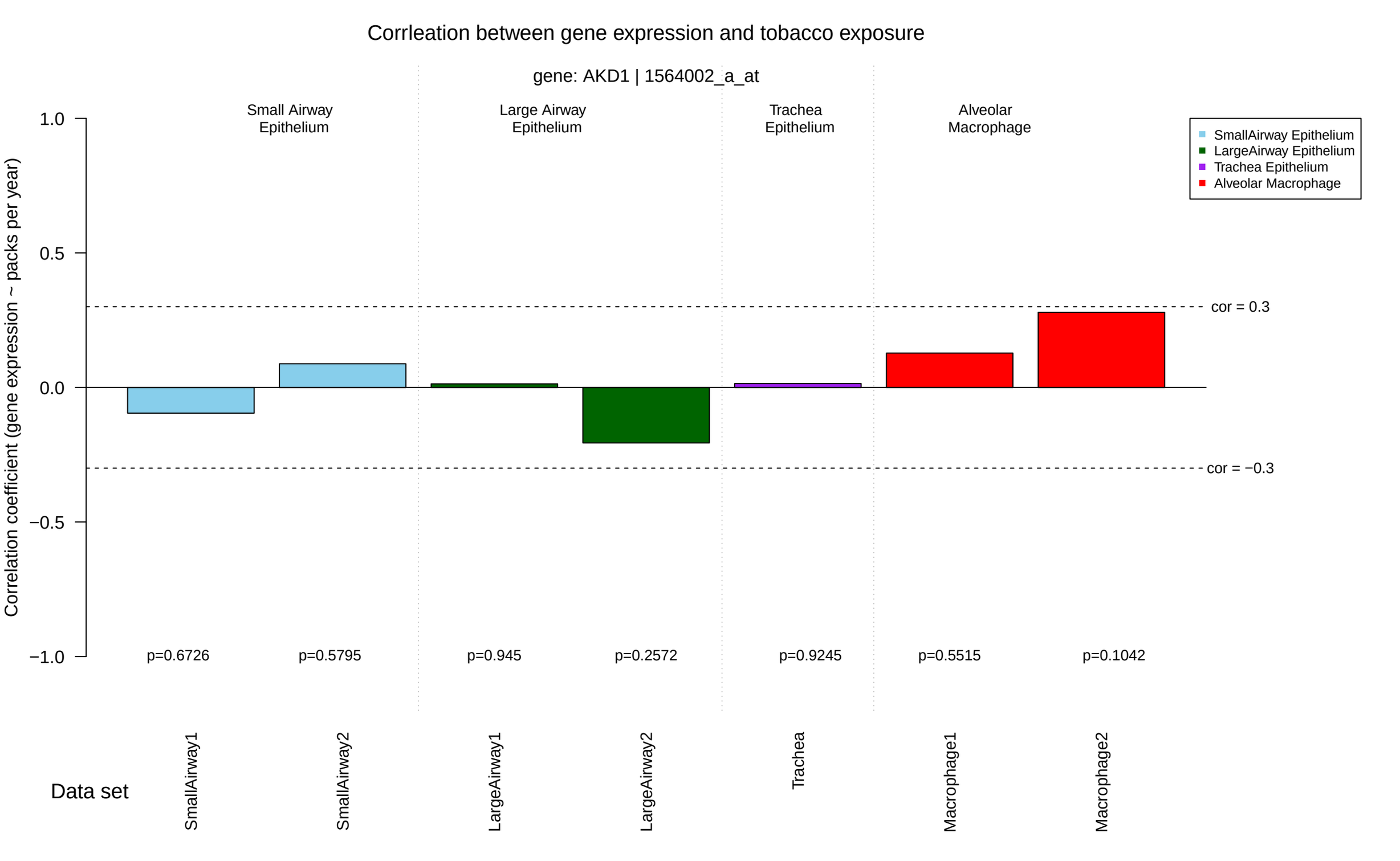

set: LargeAirway1 (p=0.6106)
AKD1 | 1564002_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1064)
AKD1 | 1564002_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9669)
AKD1 | 1564002_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.161)
AKD1 | 1564002_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0216)
AKD1 | 1564002_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKD1 | 1564002_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6726
p=0.5795
p=0.945
p=0.2572
p=0.9245
p=0.5515
p=0.1042
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2